

Semantic See-through Goggles: Linguistic Virtual Reality in (Artificial) Intelligence

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Figure 1: Semantic See-through Goggles: Linguistic Virtual Reality in (Artificial) Intelligence.

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1 Abstract

Semantic See-through Goggles are a kind of glasses through which the view is once fully verbalized and re-imaged.

Participants wear original goggles with a camera and HMD. First, a single image from the camera is converted by the AI into a single line of text. Next, an image is generated from that line of text, also by the AI. Finally, this image is presented on the display in both eyes. Both input and output are concrete images, but only the information contained in one line of text is stored, all the rest is creative. The view, in which much of the spatial positioning and material has been reworked, however, does indeed convey the meaning of the scene in front of it. These processes take place in real time, allowing the user to perceive and act on the real physical world through this redrawn view.

Every view in the goggles is a 'stereotype' of some kind, which is constructed on the basis of the specific cultural knowledge learned by the model. (For example, the view of a person looking at a 'long-haired Asian man', oscillates between a standard Anglo-Saxon, standard-bodied man and an equally Anglo-Saxon, standard-bodied woman.) This experience provides a subjective understanding of what it is like to divide the task of perceiving the world with an AI, while at the same time reminds us of the basic and inescapable fact of human perception that 'We see the world semantically'.

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